

Anchored Farey Classification and Catalan-Fibonacci Envelopes for Rational Defect Dynamics

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This paper is Paper XXII of the RIME program. It develops rational defect dynamics at fixed deficit as an independently scoped mathematical line.

Abstract

Problem. Fixed-depth finite-field route profiles do not by themselves classify the characteristic-zero state space obtained by holding defect fixed.

Approach. Rational defect dynamics is defined on primitive cusps of $\mathbb{P}^1(\mathbb{Q})$, with states represented by oriented anchored patches of the Farey tessellation.

Results. Every fixed-deficit orbit slice is finite. The reachable layer $\mathcal{C}_k$ is exactly the set of anchored Farey boundaries of cardinality k , with $|\mathcal{C}_k| = \text{Cat}_k + \text{Cat}_{k-1}$ for $k \geq 1$. Relative continuant bounds give exact coordinate, internal, global, and guarded Fibonacci envelopes. Reduction is a labelled graph isomorphism through deficit k over every finite field of characteristic $p > F_{2k+1}$.

Boundary. Exact integer certificates through deficit ten, including an independent boundary generator and a hostile $p = 23$ registry collision, support but do not replace the all-deficit proofs. The sufficient stabilization threshold is not claimed to be sharp.

Notation Table

symbol	meaning
Φ_a	labelled defect transition for $a \in \{S, R, R^{-1}\}$
$\mathcal{C}_k, \mathcal{C}_{\leq k}$	exact and cumulative reachable rational states
$\widehat{D}$	$D \cup \{\infty\}$
V_r	cusps occurring on positive Farey paths with r edges
H_k	exact-layer primitive-coordinate height
M_k^{internal}	largest determinant within one retained state
M_k^{global}	largest determinant in the cumulative cusp registry
M_k^{guard}	largest determinant after adding the one-step frontier
F_k, Cat_k	Fibonacci and Catalan numbers

1 Introduction

1.1 Problem and scope

Paper XXI studies finite-field route semantics at fixed routing depth [2]. Its low-deficit survivor layers motivate a distinct limit: this paper fixes deficit k while characteristic tends to infinity, rather than fixing depth d while field size grows. The resulting questions are whether the characteristic-zero state space is finite, whether it has an intrinsic combinatorial classification, and whether arithmetic data control finite-field stabilization. The basic objects are finite sets of primitive cusps, and the governing geometry is the Farey tessellation.

1.2 Main results

The main results are:

result	statement
Theorem 1	every cumulative rational low-deficit state space is finite and effectively enumerable
Theorem 2	reachable states are exactly anchored Farey boundaries
Corollary 3	$ \mathcal{C}_k = \text{Cat}_k + \text{Cat}_{k-1}$ for $k \geq 1$
Theorem 4	the height, internal, global, and guard determinant extrema are $F_{k+1}, F_k, F_{2k}, F_{2k+1}$
Theorem 5	characteristic $p > F_{2k+1}$ gives a labelled graph isomorphism through deficit k

The classification uses finite oriented dual patches of the Farey tessellation, explicit re-rooting, leaf attachment, and predecessor induction. Catalan counting comes from the forced cusp 1 on every nontrivial positive path. The Fibonacci result is unified by cancellation of common root words and a finite typed bridge calculation in the rooted Farey flag graph.

2 Related Work and Novelty Boundary

The Farey tessellation, its $PSL_2(\mathbb{Z})$ action, and the coding of modular orbits by continued fractions are classical; Series gives a geometric formulation in which the rational cusps are precisely the ideal vertices of the Farey tessellation [6]. Binary trees of positive rationals, mediant insertion, and their matrix encodings are also standard Stern-Brocot/Calkin-Wilf material [3, 1]. Continuants and their extremal arrangements form an established subject [3, 5], and the general Catalan convolution used in Section 4 is standard [7].

The present claims do not concern a new construction of the Farey tessellation, a new enumeration of all rationals, or a new general extremal theorem for continuants. The new surface is the typed defect system generated by the three declared maps Φ_S , Φ_R , and $\Phi_{R^{-1}}$: the exact classification of its reachable fixed-deficit states by three anchored boundary types, the separation of configuration-internal, global, and one-step guard determinant universes, and the resulting labelled finite-field graph stabilization theorem. The finite bridge calculation in Section 5 specializes standard continuant machinery to those typed registries; it is included in full because the bridge corrections are part of the theorem, not a convention inherited from the general literature.

3 Rational Defect Dynamics

3.1 Notation and exact candidate class

Write

$$\mathcal{C}_k = \{\text{reachable rational defect states of cardinality } k\}, \quad \text{Cat}_k = \frac{1}{k+1} \binom{2k}{k}.$$

The deficit-zero state is exceptional and is fixed separately:

$$\mathcal{C}_0 = \{\emptyset\}.$$

Write

$$\mathcal{C}_{\leq k} = \bigcup_{j=0}^k \mathcal{C}_j$$

for the cumulative rational state space. The finite-field analogue is

$$\mathcal{C}_{\leq k}^{(F)} = \bigcup_{j=0}^k \mathcal{C}_j^{(F)}.$$

For all rational cusp calculations, fix the canonical primitive vectors

$$v_\infty = (1, 0), \quad v_{n/d} = (n, d), \quad d > 0, \quad \gcd(|n|, d) = 1.$$

This single convention governs Farey adjacency, reduction, and every determinant universe below.

Definition 3.1 (Farey boundary).

Let B be a finite subset of $\mathbb{P}^1(\mathbb{Q})$. If its projective-line cyclic order is

$$(x_0, \dots, x_m),$$

call B a **Farey boundary** when, for the canonical primitive vectors $v_i = v_{x_i}$,

$$|\det(v_i, v_{i+1})| = 1 \quad (i \bmod m + 1).$$

For two vertices, the same Farey edge is traversed in both cyclic directions; this is the declared degenerate boundary convention. A singleton is not a Farey boundary.

For a nonempty defect state D , put

$$\widehat{D} = D \cup \{\infty\}.$$

Call D an **anchored Farey state** when $\widehat{D}$ is a Farey boundary and

$$D \cap \{0, -1\} \neq \emptyset.$$

The hatted cyclic condition is essential. The weak conditions “contains an anchor” and “successive finite cusps have determinant one” admit the infinite false family $\{0, 1/n\}$. Its closing edge from $1/n$ to infinity has determinant n and fails Definition 3.1 unless $n = 1$.

There are three disjoint marked types.

1. **zero-only**: the boundary consists of the base edge $\{0, \infty\}$ and a positive Farey path on its other side.
2. **negative-one-only**: the analogous boundary based at $\{\infty, -1\}$.
3. **both-anchors**: the boundary encloses the base ideal triangle $\{\infty, -1, 0\}$ and may extend across each of its three sides.

Lemma 3.2 (anchor-type decomposition).

Every anchored Farey state belongs to exactly one of the three marked types above. In particular, the anchor condition is not merely a requirement that one of $0, -1$ occurs: it forces the corresponding base edge or base triangle.

Proof. Consider first the case

$$D \cap \{0, -1\} = \{0\}.$$

The boundary has two arcs from ∞ to 0. Under $x \mapsto -x$, every nontrivial arc on the negative side becomes a nontrivial positive Farey path. Every such positive path contains 1: if Farey neighbors $a/b < 1 < c/d$ skipped 1, then $bc - ad = b(c - d) + d(b - a) \geq b + d \geq 2$, contrary to determinant one. Hence every nontrivial negative-side arc contains -1 . Since -1 is absent, that arc is the base edge $\{\infty, 0\}$, and the complementary arc is the positive path in the definition of zero-only.

The integral symmetry $x \mapsto -1 - x$ fixes ∞ and exchanges 0 with -1 . It gives the negative-one-only case and forces the base edge $\{\infty, -1\}$ whenever 0 is absent.

If both anchors occur, their cyclic order is $(\infty, -1, 0)$ up to reversal. The simple boundary splits into three anchor-to-anchor Farey arcs. Each arc lies on the exterior side of the corresponding edge of the base triangle; otherwise two boundary arcs would cross. Thus the base ideal triangle $\{\infty, -1, 0\}$ lies inside the patch, and each exterior arc normalizes to a positive Farey path. The three cases are disjoint by their anchor intersections and exhaustive. *QED*

3.2 Deficit transition and finite-field model

Fix the alphabet $\mathcal{A} = \{S, R, R^{-1}\}$ and the declared integral representatives

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}, \quad \widetilde{R^{-1}} = -R_{\text{alg}}^{-1} = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix},$$

where the algebraic matrix inverse is

$$R_{\text{alg}}^{-1} = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}.$$

For $a \in \mathcal{A}$, let $\tilde{a}$ denote its declared representative; write $a(x)$ for the induced projective action. Thus the alphabet label R^{-1} acts through $\widetilde{R^{-1}}$. Replacing a representative by its negative does not change the cusp map, but the displayed lifts are fixed for all determinant and bridge calculations.

For $a \in \mathcal{A}$, define

$$\Phi_a(D) = a(D \cup \{\infty\}) \setminus \{\infty\}.$$

If $\pi_a = a^{-1}(\infty)$ is the pole of a , then the modular action is injective on cusps and therefore

$$|\Phi_a(D)| = |D| + 1 - \mathbf{1}_{\{\pi_a \in D\}}.$$

This is the basic deficit monotonicity law: a transition either attaches one new cusp or re-roots at the same deficit. The primitive cusp model makes every fixed rational layer exact and discrete.

For a finite field $F = F_q$ of characteristic p , write $\mathbb{P}^1(F) = F \cup \{\infty\}$ and let $\bar{a}$ be the reduction of the declared representative $\tilde{a}$ over F . Define the finite-field transition intrinsically by

$$\Phi_{a,F}(D) = \bar{a}(D \cup \{\infty\}) \setminus \{\infty\}, \quad \mathcal{C}_k^{(F)} = \{D : |D| = k, D \text{ is reachable from } \emptyset \text{ by the } \Phi_{a,F}\}.$$

For $F = F_p$, set $\Phi_{a,p} = \Phi_{a,F_p}$ and $\mathcal{C}_k^{(p)} = \mathcal{C}_k^{(F_p)}$. Because the declared matrices have entries in the prime field and the initial cusp is ∞ , every reachable finite-field cusp lies in $\mathbb{P}^1(F_p) \subseteq \mathbb{P}^1(F_q)$. Thus the finite-field graph is defined uniformly for every finite field.

Lemma 3.3 (anchor and same-level transport).

Every nonempty successor contains 0 or -1 . If a transition preserves the deficit, then

$$\widehat{\Phi_a(D)} = a\widehat{D}.$$

Proof. The image of infinity is 0 for S, R and -1 for R^{-1} , so it is finite in every successor. If the deficit is preserved, the pole belongs to D ; hence infinity is in $a(D)$ and the hatted identity follows after removing and restoring it. *QED*

Lemma 3.4 (two-cusp rigidity).

An element of $PSL_2(\mathbb{Z})$ fixing two distinct rational cusps pointwise is the identity. Consequently, for fixed distinct rational cusps x, y and targets x', y' , at most one modular element sends (x, y) to (x', y') .

Proof. Two fixed rational eigenlines force the two eigenvalues of a determinant-one integral matrix to be rational algebraic integers with product one. They are therefore both 1 or both -1 , giving I or $-I$, the same element in $PSL_2(\mathbb{Z})$. Compose two proposed transports for the uniqueness claim. *QED*

Lemma 3.5 (finite anchored orbit slice).

For a finite cusp configuration X , let

$$\mathcal{N}(X) = \{gX : \infty \in gX, \quad gX \cap \{0, -1\} \neq \emptyset, \quad g \in PSL_2(\mathbb{Z})\}.$$

Then $\mathcal{N}(X)$ is finite and

$$|\mathcal{N}(X)| \leq 2|X|(|X| - 1).$$

Proof. For a normalized image choose a preimage of infinity, a distinct preimage of one declared anchor, and the anchor itself. There are at most $2|X|(|X| - 1)$ choices. Lemma 3.4 gives at most one group element per choice and hence at most one image. *QED*

Theorem 1 (fixed-deficit finiteness).

For every k , the cumulative reachable set $\mathcal{C}_{\leq k}$ is finite and effectively enumerable.

Proof. Induct on k . From finitely many states below level k , there are finitely many first transitions that grow to level k . After the first such transition, every remaining transition preserves size. Lemma 3.3 identifies the hatted endpoint with a modular image of the finite seed, and the anchor is present, so all such endpoints lie in the finite orbit slice of Lemma 3.5. A finite union over the seeds proves finiteness, and exact breadth-first closure terminates. *QED*

The finite-field comparison is postponed until the guarded registry has been bounded arithmetically in Section 5. Theorem 5 is stated and proved there as an explicit reduction-map graph isomorphism, rather than as an appeal to the finite-field precursor.

4 Anchored Farey Classification

Theorem 2 (anchored Farey classification).

For every $k \geq 1$,

$$D \in \mathcal{C}_k \iff D \text{ is an anchored Farey state of cardinality } k.$$

The proof is organized on the Farey tessellation rather than directly on rational-coordinate formulas.

Lemma 4.1 (boundary/patch correspondence).

Every nondegenerate Farey boundary uniquely bounds a finite connected union of Farey ideal triangles. Its dual is a finite connected subtree of the trivalent dual Farey tree. A two-vertex boundary is treated as the empty patch rooted at its declared Farey edge.

Proof. Suppose the boundary has $n \geq 3$ ideal vertices. Its cyclically ordered Farey edges form a simple ideal n -gon. The fixed Farey tessellation, rather than an abstract choice of polygon triangulation, determines the set of Farey triangles contained in this polygon. The polygon has hyperbolic area $(n-2)\pi$, while every Farey ideal triangle has area π ; their interiors are disjoint. Hence the patch contains exactly $n-2$ triangles and is finite.

The polygon interior is connected, so the adjacency dual of these triangles is connected. It has $n-2$ vertices and $n-3$ edges, one for each internal diagonal of the induced triangulation. Therefore its edge count is one less than its vertex count, and the connected dual is a tree. The contained set of Farey triangles is uniquely determined by the fixed tessellation. The two-vertex convention is the separately declared empty-patch case. *QED*

Under this correspondence:

- a zero-only state is a finite rooted patch attached on the non-base side of $\{0, \infty\}$;
- a negative-one-only state is attached on the non-base side of $\{\infty, -1\}$;
- a both-anchors state contains the base triangle and consists of three rooted side patches attached to it.

This is the patch object. Its orientation and root flag are retained data; unoriented abstract-tree isomorphism alone is too coarse.

Lemma 4.2 (transition conjugacy).

For $a \in \{S, R, R^{-1}\}$, let $\pi_a = a^{-1}(\infty)$. Exact set algebra gives

$$\widehat{\Phi_a(D)} = \begin{cases} a\widehat{D}, & \pi_a \in D, \\ a\widehat{D} \cup \{\infty\}, & \pi_a \notin D. \end{cases}$$

In the first case the modular isometry preserves the oriented dual patch and changes the flag identified with the standard rooted flag. This is a same-deficit **re-root** move.

In the second case the state has exactly one anchor. Its declared base boundary edge is carried to an edge of the new standard Farey triangle:

label	missing pole	carried boundary edge	attached triangle
R	-1	$\{\infty, 0\} \rightarrow \{0, -1\}$	$\{\infty, 0, -1\}$
S	0	$\{\infty, -1\} \rightarrow \{0, 1\}$	$\{\infty, 0, 1\}$
R^{-1}	0	$\{\infty, -1\} \rightarrow \{-1, 0\}$	$\{\infty, -1, 0\}$

Adding infinity therefore attaches one exterior Farey triangle along the carried root edge. In the dual tree this is an **attach-leaf** move. This table also shows why the poles and labels are typed data; an arbitrary modular map cannot be substituted without re-proving the root-edge statement.

Lemma 4.3 (oriented root-flag connectivity).

Fix one finite oriented dual patch P . An oriented triangle flag consists of a Farey triangle and one of its cyclically marked ideal vertices. Declare the following flags admissible:

1. all three marked flags on every triangle of P ;
2. for every boundary edge of P , the two flags on the exterior adjacent triangle whose marked vertex is an endpoint of that boundary edge.

Let f_* be the standard positively oriented flag on $(\infty, 0, -1)$ with marked vertex ∞ . The stabilizer of a positively oriented Farey triangle is its cyclic order-three rotation group. Marking one ideal vertex removes this residual stabilizer, so $PSL_2(\mathbb{Z})$ acts simply transitively on oriented triangle flags. For every flag f , let g_f be the unique element satisfying

$$g_f(f) = f_*.$$

The induced labelled action is fixed by the convention

$$\sigma_a(f) := g_f^{-1} a^{-1} f_*.$$

By simple transitivity this is the unique flag with

$$g_{\sigma_a(f)} = ag_f, \quad \widehat{D}_{\sigma_a(f)} = a\widehat{D}_f.$$

Under this convention the intuitive clockwise/counterclockwise names of R and R^{-1} may be interchanged; both labels are present, and only their connectivity is used below.

For an admissible flag define the normalized defect state

$$D_f = g_f(\partial P) \setminus \{\infty\},$$

where ∂P denotes the cusp set of the patch boundary. If f lies on an internal triangle, its other two triangle vertices show that D_f contains both 0 and -1 . If f is one of the declared exterior flags, the shared boundary edge shows that D_f contains exactly the anchor represented by its other endpoint. Conversely, a flag normalization whose boundary contains infinity and one of the two anchors is of one of these two forms. Thus

$$f \text{ is admissible} \iff D_f \cap \{0, -1\} \neq \emptyset,$$

and $D_f \cup \{\infty\}$ is the Farey boundary of the normalized patch. The graph of admissible flags under same-deficit S , R , and R^{-1} moves is connected.

Proof. With the convention above, $f' = \sigma_a(f)$ is the unique flag with $g_{f'} = ag_f$. The labels R and R^{-1} induce the two cyclic corner rotations of one triangle, while S crosses the edge incident to the corresponding oriented corner. If $\pi_a = a^{-1}(\infty)$ belongs to D_f , Lemma 4.2 gives

$$\widehat{\Phi_a(D_f)} = ag_f(\partial P) = g_{f'}(\partial P) = \widehat{D}_{f'}.$$

Thus the labelled transition is same-deficit exactly when f' remains in the admissible flag set; equivalently, exactly when the pole-membership condition holds. If f' leaves that set, Lemma 4.2 identifies the move as attach-leaf rather than a re-rooting of P .

All flags on one internal triangle are connected by the two rotations. Flags on triangles sharing an internal edge are connected by a suitable rotation, one S crossing, and another rotation. Since the internal dual subtree is connected, all internal flags are connected. Each declared exterior flag is connected across its shared boundary edge to a flag on the incident internal triangle. Hence the full admissible flag graph is connected. For the two-vertex base-edge patch, the two edge orientations are connected directly by S . QED

The exact transition census warns against a coarser statement. At $k = 5$ the whole same-level graph has four components of sizes 8, 12, 12, 24; at $k = 6$ it has six components of size 29. Connectivity is asserted only inside a fixed oriented patch class, not across all states of one deficit and not after forgetting the orientation/flag data.

Lemma 4.4 (admissible predecessor existence).

Let P be a nonempty anchored patch. There exist an anchored patch P^- with one fewer triangle, an admissible state D^- representing P^- , a label $a \in \{S, R, R^{-1}\}$, and an admissible rooting D' of P such that

$$\Phi_a(D^-) = D'.$$

Proof. If the dual tree of P has one vertex, then an admissible normalization sends P to the base ideal triangle. Choose either declared base edge, say $\{\infty, 0\}$, as the empty predecessor patch P^- . The corresponding one-anchor state D^- is reachable at deficit one, and the appropriate attach-leaf row of Lemma 4.2 adds the third vertex and produces an admissible internal rooting D' of P . This is the $k = 2$ base case.

Suppose now that the dual tree has at least two vertices. Choose a leaf triangle τ and delete it, retaining its shared parent edge e as a boundary edge of the remaining oriented patch P^- . The result is connected and is either a smaller finite patch or the declared edge-rooted empty patch; the admissible normalization below supplies its anchor marking.

View τ as the exterior triangle adjacent to e in P^- . Choose one of the two exterior flags on τ whose marked cusp is an endpoint of e . This is an admissible flag of P^- , so its standard normalization gives an anchored one-anchor state D^- . The three cyclic placements of the oriented edge e are exactly the three attach-leaf rows of Lemma 4.2. Hence one of the labels a sends D^- forward to an admissible internal rooting D' of the larger patch P , proving the displayed equality. This is a predecessor construction: geometric deletion is not a deficit-decreasing transition. *QED*

Proof of Theorem 2. The forward implication follows by induction from the two edge-rooted deficit-one states. Lemma 4.2 shows that a same-level move only re-roots the current patch and a growth move attaches one leaf, so every reachable positive-deficit state is anchored Farey.

For the reverse implication, take an anchored Farey state D . Lemma 4.1 gives its finite oriented dual patch P . If the patch is empty, D is one of the two deficit-one base states. Otherwise Lemma 4.4 supplies a smaller anchored predecessor D^- , a label a , and a rooting D' of P with $\Phi_a(D^-) = D'$. By induction D^- is reachable, hence the forward growth transition makes D' reachable. Lemma 4.3 connects D' to D by same-level moves. These moves are reversible: S is an involution, R and R^{-1} are mutual inverses, and the hatted same-level identity in Lemma 4.2 preserves the inverse pole condition. Therefore the required directed same-level path may be followed from D' to D . This proves that D is reachable without using any deficit-decreasing transition. *QED*

5 Catalan Enumeration

Let P_k be the number of positive Farey paths from 0 to infinity with k edges. The base edge gives $P_1 = 1$.

Every nontrivial positive path contains cusp 1. Indeed, if Farey neighbors $a/b < 1 < c/d$ skipped 1, then

$$bc - ad = b(c - d) + d(b - a) \geq b + d \geq 2,$$

contradicting determinant one. Splitting at 1, normalize the left and right segments by the integral determinant-one maps

$$[0, 1] \longrightarrow [0, \infty], \quad x \longmapsto \frac{x}{1 - x},$$

and

$$[1, \infty] \longrightarrow [0, \infty], \quad x \longmapsto x - 1.$$

Therefore

$$P_k = \sum_{r=1}^{k-1} P_r P_{k-r}, \quad P_1 = 1,$$

and hence

$$P_k = \text{Cat}_{k-1}.$$

The case $k = 1$ consists of the two one-anchor base edges and has no both-anchor state. For $k \geq 2$, the two one-anchor classes each contribute Cat_{k-1} . A both-anchor hatted boundary has $k + 1$ edges. Its three positive side-path edge counts are positive and sum to $k + 1$, so its count is

$$[x^{k-2}] \text{Cat}(x)^3 = \text{Cat}_k - \text{Cat}_{k-1}.$$

Corollary 3 (Catalan enumeration).

Consequently Theorem 2 gives, for $k \geq 1$,

$$|\mathcal{C}_k| = 2 \text{Cat}_{k-1} + \text{Cat}_k - \text{Cat}_{k-1} = \text{Cat}_k + \text{Cat}_{k-1}.$$

Together with $\mathcal{C}_0 = \{\emptyset\}$, this gives

$$\sum_{k \geq 0} |\mathcal{C}_k| x^k = (1+x) \text{Cat}(x) = \frac{(1+x)(1-\sqrt{1-4x})}{2x}.$$

The coefficient sequence is OEIS A167422 [4]. That cross-reference identifies the integer sequence and its algebraic generating function; it is not evidence for Theorem 2.

6 Fibonacci Envelope

6.1 Determinant universes

For a cusp x , let v_x be its declared primitive integral vector. Define

$$\mathcal{C}_{\leq k} = \bigcup_{j=0}^k \mathcal{C}_j, \quad U_{\leq k} = \bigcup_{D \in \mathcal{C}_{\leq k}} \widehat{D}.$$

The cumulative internal and global extrema are

$$M_k^{\text{internal}} = \max_{D \in \mathcal{C}_{\leq k}} \max_{x, y \in \widehat{D}} |\det(v_x, v_y)|,$$

and

$$M_k^{\text{global}} = \max_{x, y \in U_{\leq k}} |\det(v_x, v_y)|.$$

For the one-step outgoing frontier, put

$$G_k = \{\Phi_a(D) : D \in \mathcal{C}_{\leq k}, |\Phi_a(D)| = k+1, a \in \{S, R, R^{-1}\}\},$$

$$U_k^{\text{guard}} = U_{\leq k} \cup \bigcup_{E \in G_k} \widehat{E}, \quad M_k^{\text{guard}} = \max_{x, y \in U_k^{\text{guard}}} |\det(v_x, v_y)|.$$

Finally let the exact-layer coordinate height be

$$H_k = \max_{D \in \mathcal{C}_k} \max_{(n,d)=v_x, x \in D} \max\{|n|, d\}.$$

These universes are deliberately different. The $p = 23$, $k = 8$ hostile fixture shows that configuration-internal injectivity does not imply global registry injectivity, while the stabilization proof additionally needs the guard frontier to prevent a modular collision from returning an outgoing state to deficit at most k .

6.2 Root words and typed continuant bounds

Let V_r be the union of the vertex sets of all positive Farey paths from 0 to infinity having r edges. Put

$$P = I, \quad N = \widetilde{R^{-1}}, \quad M = R, \quad L = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

so that $L(x) = x/(x+1)$ and $Q(x) = x+1$. The symbol N is exactly the fixed representative from Section 2.2; no second sign convention is introduced here.

Lemma 6.1 (positive root coding and flag continuants).

(i) **Positive root coding.** Let $e_0 = (0, 1)$ and $e_\infty = (1, 0)$. Then $V_1 = \{0, \infty\}$, the registries are nested, and

$$V_{r+1} = L(V_r) \cup Q(V_r).$$

Consequently every $u \in V_r$ has a **budget- r rooted flag** $u = we_\varepsilon$, where w is a word of length exactly $r-1$ in the alphabet $\{L, Q\}$ and ε records the chosen endpoint ray. Letters in the terminal fan may fix the endpoint cusp while changing its flag; they are retained when an exact budget is required. Deleting such terminal stabilizer letters gives a minimal rooted representation of length at most $r-1$. The cusp 1 also has two endpoint flags. Thus the flag, rather than the unlabelled cusp alone, retains the endpoint and padding data used by a bridge comparison.

(ii) **Flag continuants.** Let a reduced oriented Farey flag path join two distinct endpoint rays. Partition it into maximal fans of consecutive turns in the same direction, with positive lengths $(q_1, \dots, q_t)$. Set

$$K(\emptyset) = 1, \quad K(q_1) = q_1,$$

and

$$K(q_1, \dots, q_t) = q_t K(q_1, \dots, q_{t-1}) + K(q_1, \dots, q_{t-2}).$$

Then the absolute determinant of the primitive endpoint rays is $K(q_1, \dots, q_t)$. If $m = \sum_i q_i$, then

$$K(q_1, \dots, q_t) \leq F_{m+1}.$$

Proof. For the recursion, split any positive path at its forced vertex 1. A vertex on the left or right subpath pulls back under L or Q to a vertex of a positive path with at most r edges; nesting is obtained by inserting a mediant into any edge. This proves the first inclusion. Conversely, apply L or Q to an r -edge path and use the base edge on the complementary side of 1; the resulting path has $r+1$ edges and contains the required image. The root-word statement follows by induction from the recursion.

For the determinant identity, normalize the first oriented Farey edge to the ordered primitive basis (e_∞, e_0) . Traversing a fan of q triangles replaces the new boundary ray by q times the

current ray plus the preceding ray. Thus, if D_j is the determinant from the initial ray to the ray after the j -th maximal fan, then

$$D_j = q_j D_{j-1} + D_{j-2}, \quad D_{-1} = 0, \quad D_0 = 1.$$

This is exactly the continuant recurrence. An integral determinant-one change of the initial oriented edge preserves the absolute endpoint determinant, so the identity holds for every reduced flag path. Finally, induction on the last run gives

$$qF_{n+1} + F_n \leq F_{n+q+1};$$

applying this inequality in the continuant recurrence proves $K(q_1, \dots, q_t) \leq F_{m+1}$. Alternating unit runs give equality. QED

Lemma 6.2 (all-depth relative continuant budget).

Let $r, s \geq 2$. For $u \in V_r$, $v \in V_s$, and $T, U \in \{N, M, P, L\}$, choose budget- r and budget- s rooted flags as in Lemma 6.1. If $\det(Tu, Uv)$ is nonzero, cancel the maximal common root segment and then cancel every immediate backtrack in the resulting oriented flag path. Its fan lengths $(q_1, \dots, q_t)$ satisfy

$$|\det(Tu, Uv)| = K(q_1, \dots, q_t), \quad \sum_i q_i \leq b_{T,U}(r, s),$$

where

$$b_{T,U}(r, s) = \begin{cases} r + s - 3, & T = U, \\ r + s - 1, & T \neq U, \quad T, U \in \{N, M, P\}, \\ r + s - 2, & \{T, U\} = \{P, L\}, \\ r + s, & \{T, U\} = \{N, L\} \text{ or } \{M, L\}. \end{cases}$$

If the determinant vanishes, only the resulting determinant bound is asserted; no nonzero continuant is assigned.

Proof. Let α and β be the two budgeted positive root-flag paths, of lengths $r - 1$ and $s - 1$. Terminal fan padding is chosen with the endpoint orientation used in the bridge table. After applying the typed endpoint transforms, the relative flag path is the concatenation

$$\bar{\alpha} B_{T,U} \beta, \quad B_{T,U} = T^{-1}U,$$

where the bar reverses an oriented root path. Cancelling a common root prefix or an immediate backtrack cannot increase its length. It therefore remains to compute only the finite path contribution of $B_{T,U}$ between the declared endpoint flags.

With

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

the identities $R = SQ$ and $L = SQ^{-1}S^{-1}$ hold. Direct multiplication gives, up to inversion and cyclic rotation of the base-triangle flag, four bridge classes:

typed class	representative relative matrix	endpoint-flag reduction	net correction
$T = U$	I	the two root paths meet on the same transported base edge; one terminal edge cancels	-1
distinct N, M, P	$B_{N,M}$	one branch edge remains after the corner rotation	$+1$
P, L	$B_{P,L}$	the bridge endpoint is identified with the terminal root edge	0
N, L or M, L	$B_{N,L}$ or $B_{M,L}$	two branch edges (LQ or QL) remain	$+2$

Here -1 is a forced cancellation from the two nonzero endpoint rays, not a path of negative length. The complete ordered matrix table is given in Appendix A. In particular, it verifies that no fifth typed bridge class is hidden by the flag shorthand.

Adding the two root depths to the four net corrections gives respectively

$$r + s - 3, \quad r + s - 1, \quad r + s - 2, \quad r + s.$$

Free reduction can only lower these values. Lemma 6.1 applied to the resulting reduced flag path now gives both the continuant identity and the stated budget. This argument is independent of r and s ; the finite computation is only the four-row bridge classification. **QED**

Lemma 6.3 (typed positive-Farey continuant envelope).

Let $F_0 = 0$ and $F_1 = 1$. For $r \geq 1$, every primitive vector (a, b) in V_r satisfies

$$\max\{a, b\} \leq F_r, \quad a + b \leq F_{r+1}.$$

If γ is one positive Farey path with r edges, then any two vertices x, y of γ satisfy

$$|\det(v_x, v_y)| \leq F_r.$$

For $r, s \geq 2$, the determinant envelopes between the four typed arc registries are bounded as follows:

typed pair	continuant budget	determinant bound
same type among N, M, P, L	$r + s - 3$	F_{r+s-2}
distinct types among N, M, P	$r + s - 1$	F_{r+s}
P, L	$r + s - 2$	F_{r+s-1}
N, L or M, L	at most $r + s$	F_{r+s+1}

The base cases involving V_1 are read directly from the two endpoints. The bounds F_{r+s} for $N(V_r), P(V_s)$ and F_{r+s+1} for $N(V_r), L(V_s)$ are attained by alternating Fibonacci rays.

Proof. The coordinate a , the coordinate b , and the sum $a + b$ are the absolute determinants of $v = (a, b)$ with the three primitive rays of the standard Farey triangle. A root path for $v \in V_r$ has length at most $r - 1$; the first two comparisons use that budget, while the third uses one additional base-triangle edge. Lemma 6.1 therefore gives $a, b \leq F_r$ and $a + b \leq F_{r+1}$.

Two vertices on one r -edge positive path share the part of the rooted flag path preceding their separation. Cancelling that common part leaves at most $r - 1$ fan steps, so Lemma 6.1 gives the single-path determinant bound F_r . This is not inferred merely from the two coordinate bounds.

For different typed registries, Lemma 6.2 gives the four budgets in the table; the inequality $K \leq F_{m+1}$ gives the displayed determinant bounds. For the sharp N, P and N, L rows choose the alternating rays

$$(F_r, F_{r-1}) \in V_r, \quad (F_{s-1}, F_s), (F_s, F_{s-1}) \in V_s.$$

Their determinants are F_{r+s} and F_{r+s+1} by the Fibonacci addition identities. $\square$

Lemma 6.4 (typed registry decompositions).

For every $k \geq 2$,

$$U_{\leq k} = N(V_k) \cup M(V_{k-1}) \cup P(V_k).$$

For every $k \geq 1$, the one-step guard registry is

$$U_k^{\text{guard}} = N(V_k) \cup M(V_k) \cup P(V_k) \cup L(V_k).$$

Proof. The positive registries are nested:

$$V_1 \subseteq V_2 \subseteq \dots$$

Indeed, inserting the mediant into any Farey edge refines an r -edge positive path to an $(r+1)$ -edge path without losing an old vertex.

The three arcs of the base triangle are the images under N , M , and P of the positive base arc. In a one-anchor patch the non-base path has at most k edges. In a both-anchor patch the three positive side-path budgets are positive and sum to at most $k+1$; hence the middle arc has budget at most $k-1$, while the two exterior arcs have budget at most k . Conversely, every vertex in these three registries occurs in a one-anchor path or in a three-path patch after assigning the unused side budgets to base edges. This proves the first equality.

Before same-level re-rooting, the three attach-leaf rows are:

source rooted type	missing pole	label	budget change	new contribution
zero-only	-1	R	middle $k-1 \rightarrow k$	$M(V_k)$
negative-one-only	0	S	split at the new positive pole	$L(V_k)$
negative-one-only	0	R^{-1}	exterior budgets stay at most k	already in $N(V_k) \cup P(V_k)$

The first row is the carried edge $\{\infty, 0\} \rightarrow \{0, -1\}$; the second and third are the two carried-edge choices from $\{\infty, -1\}$ in Lemma 4.2. Thus the frontier adds exactly $M(V_k)$ and $L(V_k)$ to the cumulative registry, while $N(V_k)$ and $P(V_k)$ are already present. Taking the union gives the guard decomposition.

The converse inclusions are constructive:

target registry	source patch	side budgets	operation
$N(V_k)$	one-anchor normalized patch	one non-base side $\leq k$	same-level rooting
$P(V_k)$	reflected one-anchor patch	one non-base side $\leq k$	same-level rooting
$M(V_{k-1})$	both-anchor patch	middle $k-1$, exterior $1, 1$	same-level rooting
guard $M(V_k)$	zero-only source	middle $k-1 \rightarrow k$	R growth row
guard $L(V_k)$	negative-one-only source	positive split at pole 0	S growth row

For $N(V_k)$ and $P(V_k)$, choose the corresponding one-anchor positive path. For $M(V_{k-1})$, place the path on the middle side of the base triangle and take the two exterior side budgets to

be one; the total deficit is at most k . Unused budget is inserted by mediants, using the nesting of the V_r . The two guard rows provide an explicit source state and label for every new $M(V_k)$ or $L(V_k)$ cusp, while $N(V_k)$ and $P(V_k)$ are already cumulative. Thus both displayed unions are equalities rather than only upper bounds. QED

Theorem 4 (unified Fibonacci envelope).

For every $k \geq 1$,

$$H_k = F_{k+1}, \quad M_k^{\text{internal}} = F_k, \quad M_k^{\text{global}} = F_{2k}, \quad M_k^{\text{guard}} = F_{2k+1}.$$

The proof uses the single relative budget lemma above, rather than three separate extremal arguments.

Proof. For $k = 1$, the four equalities follow directly from the two base states and their first guard frontier. Assume henceforth that $k \geq 2$. Lemma 6.3 and the first registry decomposition of Lemma 6.4 give $H_k \leq F_{k+1}$. Equality is attained by

$$x_k = N(F_k/F_{k-1}) = -F_{k+1}/F_k.$$

This cusp occurs in an exact deficit- k one-anchor Fibonacci-comb state, so it is a witness for the exact-layer height rather than only for the cumulative registry.

For one-anchor states, two cusps lie on one k -edge positive path, so Lemma 6.3 gives the internal bound F_k . For a both-anchor state, two cusps on different side paths of budgets r, s satisfy $r + s \leq k$, because the third side has positive budget. The distinct-arc clause again gives F_k . Equality is attained in the positive Fibonacci comb by the pair $0, F_k/F_{k-1}$.

For the global registry, Lemma 6.4 gives exterior budgets k, k and middle budget $k - 1$. The largest typed bound in Lemma 6.3 is therefore F_{2k} , for the N, P pair. It is attained by

$$x_k = (-F_{k+1}, F_k), \quad y_k = (F_{k-1}, F_k),$$

because

$$|\det(x_k, y_k)| = F_k(F_{k+1} + F_{k-1}) = F_{2k}.$$

Finally, the guard decomposition has four budget- k typed registries. Lemma 6.3 bounds every pair by F_{2k+1} ; the only extra bridge budget is the N, L or M, L case. The N, L pair

$$x_k = (-F_{k+1}, F_k), \quad z_k = (F_k, F_{k+1})$$

attains the bound since

$$|\det(x_k, z_k)| = F_{k+1}^2 + F_k^2 = F_{2k+1}.$$

This completes the $k \geq 2$ case. QED

7 Finite-Field Stabilization

Theorem 5 (explicit finite-field stabilization).

Set $\Phi_{a, \mathbb{Q}} = \Phi_a$ and $\mathcal{C}_{\leq k}^{(\mathbb{Q})} = \mathcal{C}_{\leq k}$. For $K \in \{\mathbb{Q}, F_q\}$, define the labelled graph through deficit k by

$$\mathcal{G}_{\leq k}^{(K)} = \left\{ D \xrightarrow{a} \Phi_{a, K}(D) : D, \Phi_{a, K}(D) \in \mathcal{C}_{\leq k}^{(K)}, a \in \mathcal{A} \right\}.$$

The one-step guard frontier is not part of this target graph; it is retained only to rule out an out-of-range rational successor folding back into the in-range finite-field graph.

Let

$$\Delta_k = \max\{k, M_k^{\text{guard}}\} = \max\{k, F_{2k+1}\} = F_{2k+1} \quad (k \geq 1).$$

For every finite field F_q of characteristic $p > \Delta_k$, reduction of primitive rational cusps induces a labelled graph isomorphism

$$\text{red}_p : \mathcal{G}_{\leq k}^{(\mathbb{Q})} \xrightarrow{\cong} \mathcal{G}_{\leq k}^{(F_q)},$$

where the right-hand graph uses the intrinsically defined transitions from Section 2.2. In particular,

$$|\mathcal{C}_j^{(F_q)}| = |\mathcal{C}_j(\mathbb{Q})| \quad (0 \leq j \leq k).$$

Proof. By Theorem 4, every pair of distinct cusps in the guarded registry has nonzero determinant of absolute value at most F_{2k+1} . Hence reduction modulo p is injective on the entire registry, including infinity. Integral matrix multiplication commutes with reduction, so each rational labelled edge reduces to the corresponding finite-field edge with the same cardinality.

For surjectivity, induct along a finite-field path that stays through deficit k . Lift its current state by induction. Its next rational successor either has size at most k , or is an element of the one-step guard frontier and has size $k+1$. The latter remains injective after reduction and therefore cannot appear as an in-range finite-field state. Thus every in-range finite-field successor has a unique rational lift. Finally, the prime-field observation in Section 2.2 identifies the same graph inside every F_q of characteristic p . QED

8 Exact Finite Certificates

8.1 Low-deficit stability certificate

The exact integer producer closes the rational transition system through deficit eight with layer counts

$$(|\mathcal{C}_0|, \dots, |\mathcal{C}_8|) = (1, 2, 3, 7, 19, 56, 174, 561, 1859).$$

The cumulative registry contains 320 primitive cusps and has maximum global determinant 987; the one-step guarded registry contains 448 cusps and has maximum determinant 1597. The $p = 23$ hostile fixture exhibits a genuine global collision (2/5 and 5, determinant -23) despite configuration-local injectivity. This is why Theorem 5 uses the guarded registry.

8.2 Deficit-ten Farey-Catalan certificate

An independent anchored-boundary generator and the transition closure agree exactly through $k = 10$:

$$(|\mathcal{C}_0|, \dots, |\mathcal{C}_5|) = (1, 2, 3, 7, 19, 56),$$

$$(|\mathcal{C}_6|, \dots, |\mathcal{C}_{10}|) = (174, 561, 1859, 6292, 21658).$$

The category counts agree with $|\mathcal{C}_k| = \text{Cat}_k + \text{Cat}_{k-1}$. The same certificate checks the four Fibonacci extrema and the hostile same-level component profiles at $k = 5$ and $k = 6$.

9 Claim Status and Boundary

surface	reader-facing status	evidence basis
fixed-deficit finiteness	Theorem	Theorem 1
anchored Farey classification	Theorem	Theorem 2
Catalan enumeration	Theorem	Corollary 3
four Fibonacci envelopes	Theorem	Theorem 4
guarded finite-field stabilization	Theorem	Theorem 5
exact layers through deficit ten	Computational Certificate	integer replay and local closure receipt
hostile $p = 23$ registry collision	Computational Certificate	exact determinant and finite-field replay
sharp thresholds, minimal quotient, and joint (d, k) asymptotics	Research Program	open

The theorem statements are conditional on the declared modular generators, primitive cusp convention, anchored state definition, and guarded registry. The finite certificates use integer-exact arithmetic and establish their recorded finite scopes only. Their transition-closure and anchored-boundary generators share arithmetic primitives but not membership logic. A local receipt binds and replays the declared bytes; it does not independently validate its own validator or promote a sufficient threshold to a sharp one.

10 Conclusion and Open Problems

The rational defect dynamics has a finite fixed-deficit model with an exact geometric classification. Anchored Farey patches give Catalan state counts, and typed continuant budgets give Fibonacci arithmetic envelopes. The guarded determinant registry supplies a concrete, if generally non-sharp, stabilization threshold for reduction to large finite fields.

Open problems include sharp thresholds, a canonical minimal quotient of the oriented patch transition system, uniform asymptotics in k , and extensions to other arithmetic Fuchsian carriers. Route-depth and fixed-deficit limits remain complementary rather than mutually determining.

Appendix A Complete finite typed bridge table

The finite typed part of Lemma 6.2 can be checked without geometric shorthand. For $u = (a, b)$ and $v = (c, d)$, the following table lists every unordered typed pair. The matrix column is $T^{-1}U$; the normal form is the signed determinant before taking its absolute value.

pair	bridge	reduced flag class	correction	signed determinant
N, N	I	terminal cancellation	-1	$ad - bc$
M, M	I	terminal cancellation	-1	$ad - bc$
P, P	I	terminal cancellation	-1	$ad - bc$
L, L	I	terminal cancellation	-1	$ad - bc$
N, M	$B_{N,M}$	one branch edge	$+1$	$-ac - bc - bd$
N, P	$B_{N,P}$	one branch edge	$+1$	$-ac - ad - bd$
M, P	$B_{M,P}$	one branch edge	$+1$	$-ac - bc - bd$
N, L	$B_{N,L}$	two branch edges	$+2$	$-2ac - ad - bc - bd$
M, L	$B_{M,L}$	two branch edges	$+2$	$-ac - 2bc - bd$
P, L	$B_{P,L}$	endpoint identified	0	$ac + ad - bc$

The nonidentity bridge matrices are

$$\begin{aligned} B_{N,M} = B_{M,P} &= \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}, & B_{N,P} &= \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}, \\ B_{N,L} &= \begin{pmatrix} 1 & 1 \\ -2 & -1 \end{pmatrix}, & B_{M,L} &= \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}, \\ B_{P,L} &= \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}. \end{aligned}$$

The reversed ordered pair has the same absolute determinant. The N, M, P rows are related by cyclic corner rotations; the L rows are the two exterior-edge cases. Hence this table exhausts $\{N, M, P, L\}^2$ modulo symmetry, and the only unbounded part of the bridge calculation is the common L/Q root word handled by the continuant recurrence.

Appendix B Computational Artifacts

The paper-owned sources and source-addressed evidence are available in the [RIME repository](#). The paper-owned directory is `experiments/paper22/`; paths below are relative to it.

role	repository path
rational transition producer	<code>exact_rational_low_deficit.py</code>
Farey/Catalan producer	<code>farey_catalan_structure.py</code>
exact result records	<code>results/</code>
validators	<code>validation/</code>
evidence manifest	<code>evidence-manifest.json</code>

The validation receipts exclude themselves from their closures. Their status is local exact replay and source-closure verification, not independent mathematical validation.

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